# EVALUATION OF NLP SYSTEMS

## Lecture 5 - Metrics & Measurements

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- context
- examples

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# Different approaches for each input-output relation

< This is an example.
> DT VBZ DT NN .

## one output per input

POS, NER, WSD

< What's in this picture?
> This is a brown dog.

## text output for each input

dialogue, machine translation, image captioning

*for a set of inputs, to what extent does system output correspond to outputs that are correct or desirable?*

< search: evaluation
> <doc1>, <doc2>, <doc3>

## multiple outputs per input

document retrieval, related books suggestions

< This is an example.
> (ROOT (S (NP (DT This)) (VP (VBZ is) (NP (DT an) (NN example))) (. .)))

## structured outputs

syntactic parsing, semantic parsing

< 'I'm studying NLP.'
> Positiveness: 97.45%

## output values on a scale

sentiment analysis, automatic grading

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Resnik & Lin (2010))

needed to measure performance and quantify correctness and quality

evaluation metrics are performance indicators

help detect problems and compare systems

metrics can be used for automatic evaluation

correlation with human judgment is a desirable property

sometimes they are proxies for what we really want to measure

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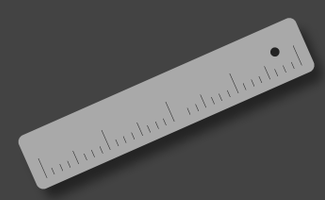
# METRICS AREN'T DECREES!

- experts propose metrics
- some are accepted by the community
- people start adopting them
- they become popular and notorious
- it feels like they were always there…

- but don't use them *only* because of tradition
- they sometimes need to be adapted or customized
- or eventually replaced
- (although standards are important e.g. for comparison)



each metric has its own range, utility, interpretation, use cases, limitations and scope

**ALWAYS THINK: IS THIS METRIC SUITABLE TO EVALUATE MY SYSTEM?**

Smatch: an Evaluation Metric for Semantic Feature Structures

CobaltF: A Fluent Metric for MT Evaluation

A Resource-Free Evaluation Metric for Cross-Lingual Word Embeddings Based on Graph Modularity

DSCORER: A Fast Evaluation Metric for Discourse Representation Structure Parsing

ReVal: A Simple and Effective Machine Translation Evaluation Metric Based on Recurrent Neural Networks

BLEU: a Method for Automatic Evaluation of Machine Translation

PEM: A Paraphrase Evaluation Metric Exploiting Parallel Texts

ParaMetric: An Automatic Evaluation Metric for Paraphrasing

SEMBLEU: A Robust Metric for AMR Parsing Evaluation

# METRICS ARE BORN ALL THE TIME

let's check out some of the currently most common in NLP

MAXSIM: A Maximum Similarity Metric for Machine Translation Evaluation

METEOR: An Automatic Metric for MT Evaluation with High Levels of Correlation with Human Judgments

EMMA: A Novel *E*valuation *M*etric for *M*orphological *A*nalysis

LAYERED: Metric for Machine Translation Evaluation

Edit Distance: A Metric for Machine Translation Evaluation

ETER: a new metric for the evaluation of hierarchical named entity recognition

Source: ACL Anthology

# Confusion matrix

visualization of the performance of a classifier

*sum of main diagonal is number of total correct predictions*

*sum of first column is the number of times class A was predicted*

*this cell means that class A was wrongly classified as E three times*

| | | prediction | | | | |
|---|---|---|---|---|---|---|
| | | class A | class B | class C | class D | class E |
| gold | class A | 10 | 1 | 3 | 4 | 3 |
| | class B | 2 | 24 | 5 | 2 | 2 |
| | class C | 0 | 2 | 32 | 1 | 0 |
| | class D | 1 | 5 | 7 | 25 | 0 |
| | class E | 0 | 3 | 1 | 4 | 39 |

*sum of last row is how many times class E appears in gold standard*

*sum of all cells is the total number of datapoints*

*this cell means that class C was correctly classified 32 times*

# Accuracy / Precision / Recall / F1 Score

common in classification tasks

| | | prediction | |
|---|---|---|---|
| | | TRUE | FALSE |
| gold | TRUE | True Positive | False Negative |
| | FALSE | False Positive | True Negative |

*what if umbalanced set?*

$$accuracy = \frac{correct}{total} = \frac{TP + TN}{TP + FP + FN + TN}$$

*what if output one TRUE and all FALSE?*

$$precision = \frac{correct_true}{predicted_true} = \frac{TP}{TP + FP}$$

all in [0, 1]
the higher, the better

$$recall = \frac{correct_true}{true_true} = \frac{TP}{TP + FN}$$

*what if output all TRUE?*

$$F1 = \frac{(\beta^2 + 1) * precision * recall}{(\beta * precision) + recall}$$

*trying to find a compromise!*

# Accuracy / Precision / Recall / F1 Score

*when there are more than two classes, evaluate precision, recall and F1 for each of them*

| | | prediction | | |
|---|---|---|---|---|
| | | class A | class B | class C |
| gold | class A | aa | ab | ac |
| | class B | ba | bb | bc |
| | class C | ca | cb | cc |

$$accuracy = \frac{aa + bb + cc}{aa + ab + ac + ba + bb + bc + ca + cb + cc}$$

$$precision_A = \frac{aa}{aa + ba + ca} \qquad precision_B = \frac{bb}{ab + bb + cb} \qquad precision_C = \frac{cc}{ac + bc + cc}$$

$$recall_A = \frac{aa}{aa + ab + ac} \qquad recall_B = \frac{bb}{ba + bb + bc} \qquad recall_C = \frac{cc}{ca + cb + cc}$$

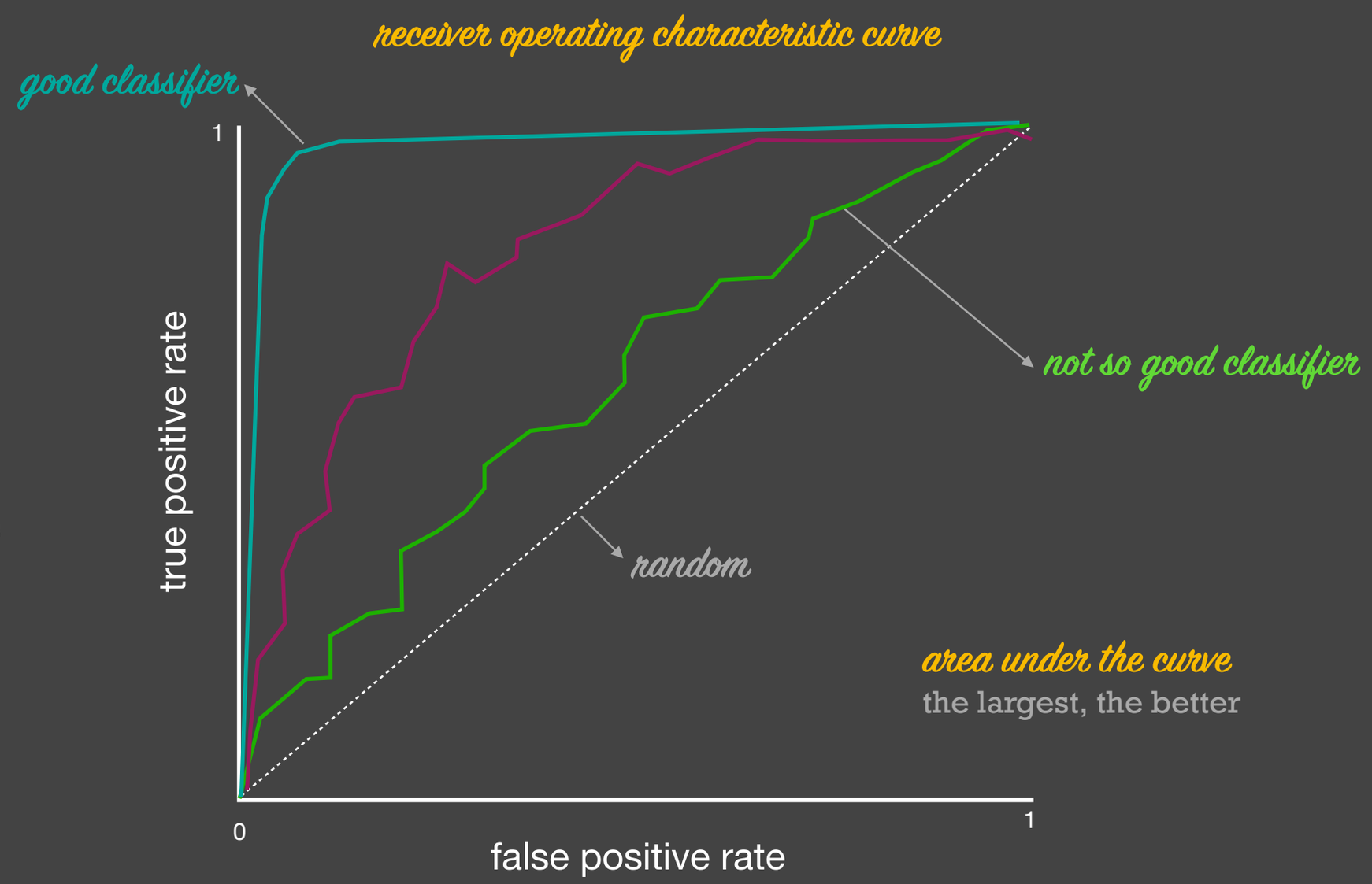
F1 Score can also be computed per class. Some variants: macro-averaged, micro-averaged or weighted.

# ROC / AUC

love: 0.97
hate: 0.12
happiness: 0.89
sadness: 0.23
patience: 0.62
perseverance: 0.57
ambition: 0.49
sincerity: 0.51

if your classifier scores words like that, what's the threshold between positive and negative words? 0.5? 0.6? 0.45? 0.7?

verification of the diagnostic ability of a binary classifier for varying thresholds



# Perplexity / Entropy / Cross-entropy / KL Divergence

common in language modelling

inverse probability of the test set W, normalized by the number N of items

*perplexity*

$$PP(W) = P(w_1, w_2, \ldots, w_N)^{-\frac{1}{N}} = \sqrt[N]{\frac{1}{w_1, w_2, \ldots, w_N}}$$

lower perplexity means the model is better at predicting the data

$$PP = 2^{H(p,q)}$$

*entropy*

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log P(x_i)$$

lower entropy means less uncertainty

*cross-entropy*

$$H(p, q) = -\sum_{i=1}^{n} P(x_i) \log Q(x_i)$$

common loss function in neural networks

*Kullback-Leibler divergence*

$$D_{KL}(p \| q) = -\sum_{i=1}^{n} P(x_i) \log \frac{Q(x_i)}{P(x_i)}$$

measures how much one probability distribution differs from another one

$$H(p, q) = H(p) + D_{KL}(p \| q)$$

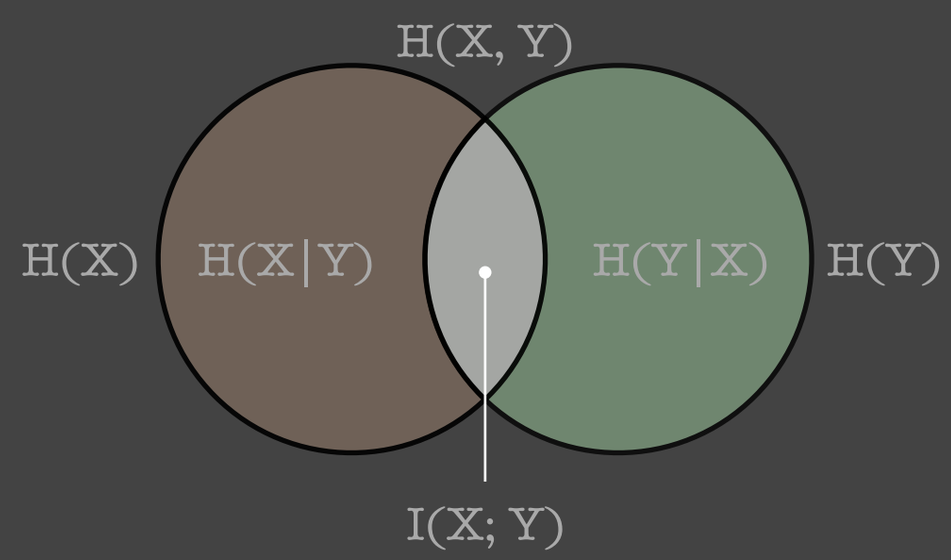
*surprisal*

$$I(x) = -\log P(x)$$

seeings events (e.g. words) with lower probability makes us more 'surprised'

# Other metrics from Information Theory

useful for associations between words or characters




joint entropy,
conditional entropy

*mutual information*

$$I(X;Y) = \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} p_{(X,Y)}(x,y) \boxed{\log \left( \frac{p_{(X,Y)}(x,y)}{p_X(x)\, p_Y(y)} \right)}$$

how much knowing one variable reduces uncertainty about another

expected value of pmi

I(X; Y) = 0 if x and y are independent

*pointwise mutual information*

$$pmi(x;y) \equiv \log \frac{p(x,y)}{p(x)p(y)} = \log \frac{p(x\,|\,y)}{p(x)} = \log \frac{p(y\,|\,x)}{p(y)}$$

# Similarity

common in semantics

*Jaccard Index*

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

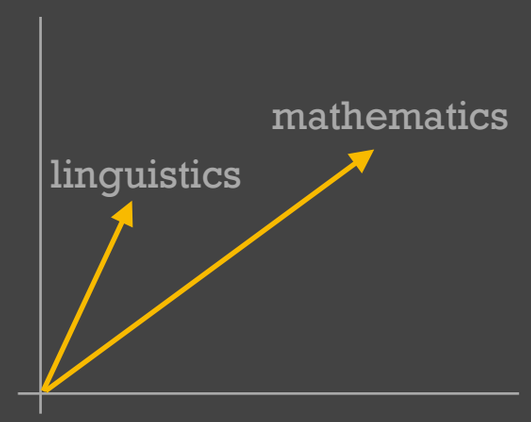
*Dice's coefficient*

$$DC = \frac{2|X \cap Y|}{|X| + |Y|}$$

*cosine similarity* for continuous representations

$$cosSim = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

-1: exactly opposite
0: orthogonal
1: exactly the same



# TF-IDF

common in information retrieval

*term frequency*

$$tf(t,d) = n_{t,d}$$

how many times term *t* appears in document *d*

many alternatives: 1 is term *t* appears in document *d* else 0
augmented frequency
adjusted for document length

*inverse document frequency*

$$idf_t = \log \frac{N}{df_t}$$

total number of documents

how many documents contain term *t*

*tf-idf*

$$\textbf{tf-idf}(t,d,D) = tf(t,d) * idf(t,D)$$

usually comes with weights

# Word Error Rate / Edit Distance

common in speech recognition

*word error rate*

e.g. for sentences

$$WER = \frac{S + D + I}{N}$$

substitutions + deletions + insertions

number of words in reference text

*Levenshtein distance*

e.g. for strings of carachters

$$\mathrm{lev}_{a,b}(i,j) = \begin{cases} \max(i,j) & \textbf{if } \min(i,j) = 0 \\ \min \begin{cases} \mathrm{lev}_{a,b}(i-1,j) + 1 \\ \mathrm{lev}_{a,b}(i,j-1) + 1 \\ \mathrm{lev}_{a,b}(i-1,j-1) + 1_{(a_i \neq b_j)} \end{cases} & \textbf{otherwise} \end{cases}$$

# BLEU & Co.

common in machine translation and text summarization

## BLEU

Papineni et al. (2002)

$$p_n = \frac{\sum_{C \in \{Candidates\}} \sum_{n\text{-}gram \in C} Count_{clip}(n\text{-}gram)}{\sum_{C' \in \{Candidates\}} \sum_{n\text{-}gram' \in C'} Count(n\text{-}gram')}$$

$$\mathrm{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

c: length of candidate translation
r: reference corpus length
p: modified precision score

'modified unigram precision'

a measure of n-gram matches between a reference text and candidate translations

'brevity penalty'

help avoid translations that are too short

many adaptations, e.g.

## ROUGE

Lin (2004)

ROUGE-N, ROUGE-L, ROUGE-W, ROUGE-S, ROUGE-SU

## METEOR

Banerjee & Lavie (2005)

# Mean Squared Error

a regression loss function

*difference between the model's prediction and the 'true' value*

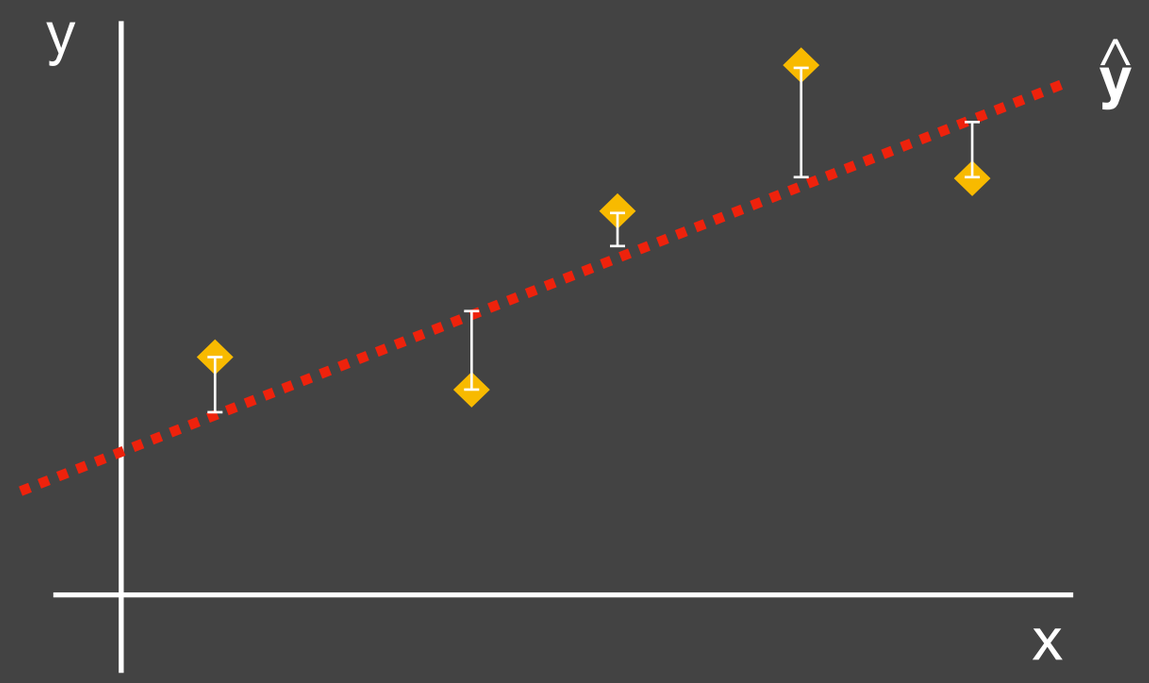
$$MSE = \boxed{\frac{1}{n}\sum_{i=1}^{n}}\left(\boxed{Y_i - \hat{Y}_i}\right)^{\boxed{2}}$$

*square it to avoid negative and positive differences cancelling each other*

*take the mean over all n datapoints*

minimum is 0

we'd like it to be as low as possible



# Cluster analysis

## how good are the system cluster outputs

*Rand Index* [0, 1]

Given a set of $n$ elements $S = \{o_1, \ldots, o_n\}$ and two partitions of $S$ to compare, $X = \{X_1, \ldots, X_r\}$, a partition of $S$ into $r$ subsets, and $Y = \{Y_1, \ldots, Y_s\}$, a partition of $S$ into $s$ subsets, define the following:

- $a$, the number of pairs of elements in $S$ that are in the same subset in $X$ and in the same subset in $Y$
- $b$, the number of pairs of elements in $S$ that are in different subsets in $X$ and in different subsets in $Y$
- $c$, the number of pairs of elements in $S$ that are in the same subset in $X$ and in different subsets in $Y$
- $d$, the number of pairs of elements in $S$ that are in different subsets in $X$ and in the same subset in $Y$

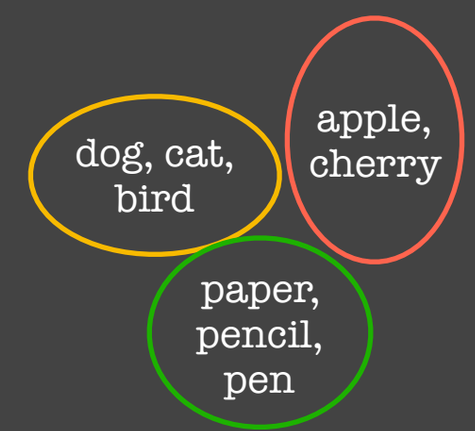
Source: Wikipedia

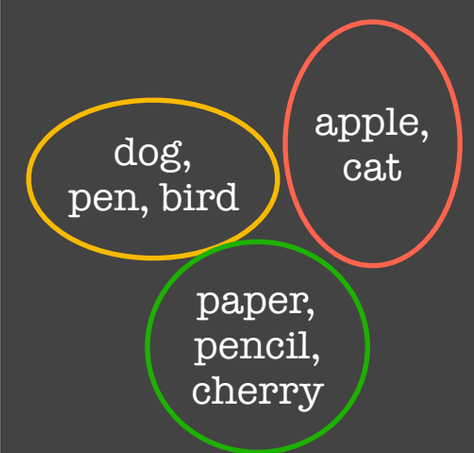
$$R = \frac{a+b}{a+b+c+d} = \frac{a+b}{\binom{n}{2}}$$

*Adjusted Rand Index* corrected for chance [-1, 1]

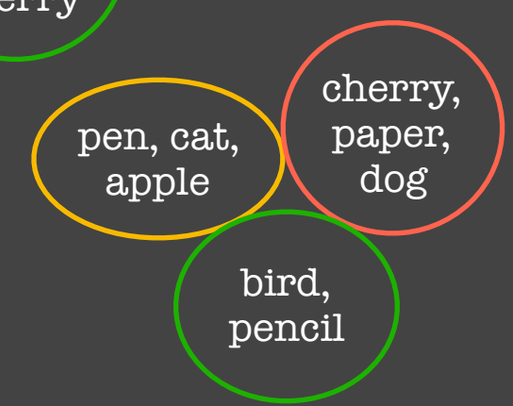
$$ARI = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] \Big/ \binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] \Big/ \binom{n}{2}}$$

gold standard clusters



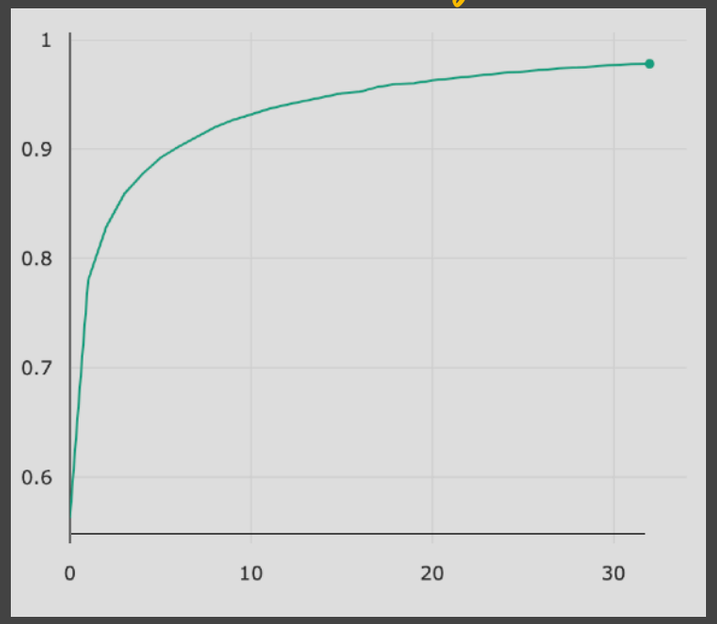

which output is better?



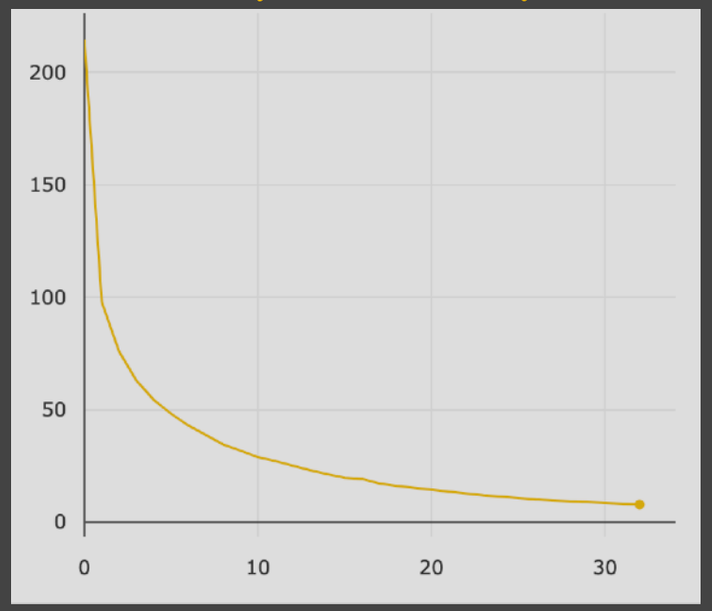
# Learning curves

keeping track of how a metric evolves over time

*F1 Score (training set)*

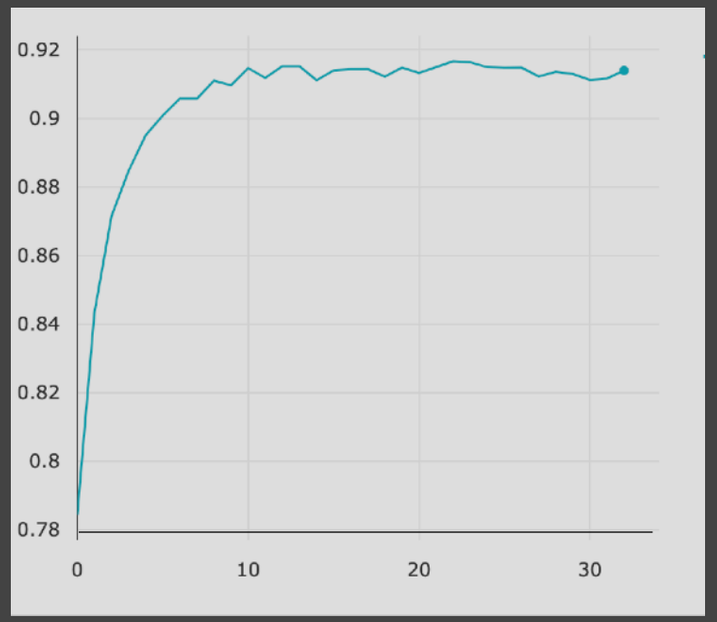



*Cross-entropy loss (training set)*

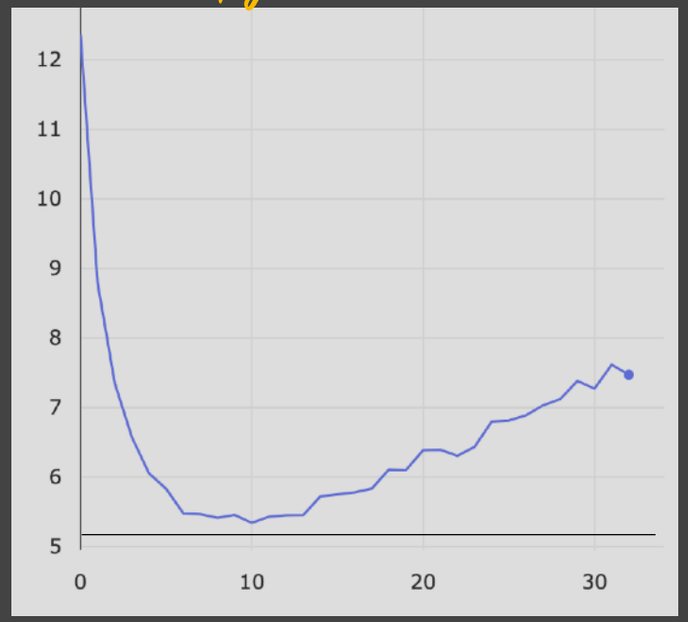



used to examine underfitting and overfitting

*F1 Score (validation set)*




*Cross-entropy loss (validation set)*




optimizing cross-entropy as a proxy for the metrics we'd like to see increasing (F1)

# Human evaluation

## freedom to ask all sorts of subjective questions

### Likert scale

*This is a coherent text.*
*The answer was helpful.*
*This sounds like a human.*
*The sentence is correct.*

### ranking

*Order sentences from most coherent (1) to less coherent (3).*

*[ 3 ] <output system A>*
*[ 1 ] <output system B>*
*[ 2 ] <output system C>*

### preference

*Which sentence sounds more formal?*

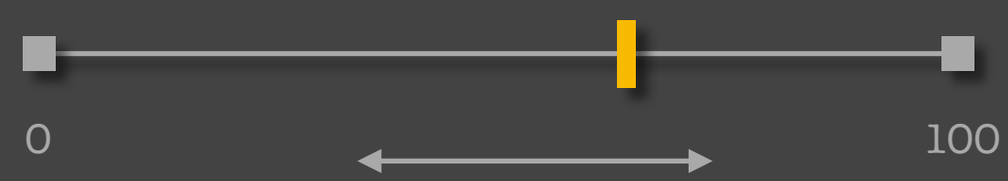
*[ X ]<output system A>*
*[ ] <output system B>*

### rating

*How satisfied are you with the search results?*

### continuous score

*From 0 to 100, how much did you enjoy talking to the virtual assistant?*




answers alone are not enough, a robust statistical analysis is needed!

e.g. Student's T test, Mann-Whitney U test, McNemar's test

# Many others

statistics, software metrics, algorithmic efficiency, business analytics…

*confidence intervals* *mean* *log-likelihood* *fallback rate*

*covariance* *p-value* *standard deviations*

*Spearman's ρ* *variance*

*purity* *precision@K* *Fowlkes–Mallows index*

*user satisfaction* *Hopkins statistic*

*click-through rate*

*response time* *median* $R^2$

*mean absolute error* *correlation*

*time & memory use* *engagement*

*specificity*

*goal completion rate*

*homogeneity score* *V measure* *Pearson correlation*

*(un)labelled attachment score*

for each new project, check the literature to get up-to-date with current techniques!

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# See you next week!

- ☑ context
- ☑ examples

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